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Arc and keyhole behavior in narrow-gap oscillating laser-MIG hybrid welding of thick aluminum alloys

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Highlights:

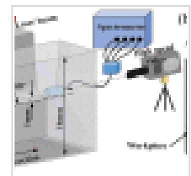
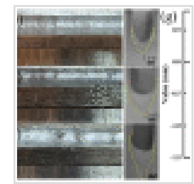
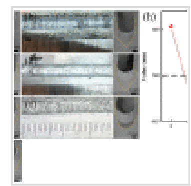
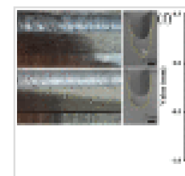
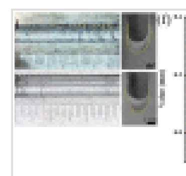
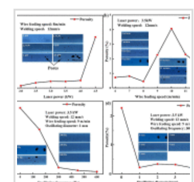
Outline

- Narrow gap oscillating laser-MIG hybrid welding of aluminum alloys was conducted.
- Effects of oscillating laser on weld formation and porosity were illustrated.
- Relationship of weld formation and arc behavior in narrow gap was clarified.
- Formation of bulge was suppressed and keyhole stability was improved.

Abstract

Narrow-gap oscillating laser-MIG hybrid welding was developed to weld thick aluminum alloys. Weld formation and porosity defect with various oscillating laser parameters were systematically analyzed. Furthermore, the corresponding influencing factors, i.e., arc and keyhole dynamic behaviors were discussed. The difference between both sidewall penetrations was small while wire feeding speed was high and oscillating laser diameter was lower than 3 mm. When laser oscillating frequency was higher than 300 Hz, porosity was less than 0.5%. Arc burnt stably and the deflection angles were small while laser oscillating frequency was 300–450 Hz and diameter was 1–2 mm. The small fluctuation of arc resulted in small difference between both sidewall penetrations. With the action of oscillating laser, the charged particles increased and diffused to the upper part of the groove. Therefore, arc was attracted by the charged particles and struck at the center bottom of groove. While the arc burnt stably, the difference between both sidewall penetrations was small. The pore formation was significantly inhibited by oscillating laser, as the keyhole opening size increased and fluctuation degree decreased. With the action of oscillating laser, the ratio of keyhole opening width to depth increased, the probability of Rayleigh jet instability on the keyhole wall was reduced.

Figures (18)

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Tables (3)

Keywords

Narrow-gap; Oscillating laser-MIG hybrid welding; Arc behavior; Keyhole dynamic behavior

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1. Introduction

Prized for the high specific strength and excellent corrosion resistance [1], [2], thick aluminum alloys exhibit a great application prospect in high-speed train, aerospace, high pressure vessels, shipbuilding industry and so on [3], [4]. However, the welding of thick aluminum alloys is the key technology for structure manufacturing in the above-mentioned fields. The amount of filler metal is reduced, the welding efficiency is improved and the welding heat input is further reduced by utilizing narrow-gap groove, thereby the joint softening and deformation were inhibited [5], [6], [7]. Compared with arc welding and laser wire filling welding, laser-arc hybrid welding possesses the advantages of low welding heat input, high welding efficiency and good process adaptability [8], [9], [10]. Therefore, the narrow-gap laser-MIG hybrid welding (NGLMHW) exhibits strong application prospects for welding of thick aluminum alloys.

Arc and keyhole dynamic behaviors were complicated during the laser-arc hybrid welding process of aluminum alloys [11], [12], [13], [14]. Jia [15] *et al.* pointed out that arc behavior was influenced by the pulsed laser frequency. The arc was compressed and burnt stably, when the continuity of pulsed laser was enough to evaporate the base material to form plasma. Li [16] *et al.* found that the keyhole was more likely to collapse owing to the greater impact of droplet on the keyhole when it was close to the laser irradiation center. Norris *et al.* [17] demonstrated that the stability of keyhole was influenced by weld pool geometry and welding speed during laser welding process. Huang *et al.* [18] concluded that collapse frequency of keyhole wall decreased with the increase of welding speed, as higher welding speed reduced keyhole depth which could promote the stability of keyhole. Besides, researchers indicated that keyhole induced porosity was related to the collapse of keyhole wall and affected the mechanical properties of joint seriously [19], [20], [21].

During narrow-gap laser-arc hybrid welding process, the arc behavior is affected by the sidewall, which in turn affects weld formation. Meng *et al.* [22] pointed out that the weld formation and laser induced plasma were influenced by groove parameters. Uneven weld was formed in narrow groove, as arc was attracted by sidewalls and droplet was transferred to the sidewalls. Yu *et al.* [23] concluded that an inclined weld was produced, since the droplet was transferred towards the molten pool along the arc deflection direction when the arc deflected to the sidewall. Meng *et al.* [24] suggested that the lack of fusion occurred in next pass when the groove cannot be fully filled by previous pass. Besides, the lack of fusion defect was effectively inhibited by a concave shape of the previous pass. To suppress arc deflection, Cai *et al.* [25] introduced an alternating magnetic field into narrow-gap laser-MIG welding. The laser-induced plasma was weakened as the magnetic field intensity increased, and the arc remained in a magnetic restraint state. The phenomenon of arc climbing along the inner wall in narrow gap was effectively inhibited.

Numerous researchers were devoted to stabilizing the arc and keyhole behavior, improving the weld quality of NGLMHW by optimizing welding processing parameters, applying oscillating laser and so on. Cai *et al.* [26] pointed out that unstable droplet transfer behavior was attributed to unstable and high velocity shielding gas flow. Wang *et al.* [27] indicated that arc behavior was influenced by the melted metal at wire tip. The poor formation of weld in narrow-gap groove was attributed to unstable behavior of melted metal and resultant arc deflection. Yu *et al.* [28] found that the arc deflection was suppressed efficiently by the oscillating laser during narrow-gap laser-MIG hybrid welding of titanium alloys. Besides, the keyhole stability was enhanced, since the keyhole size was enlarged and the impact action by droplet was cushioned. Narrow-gap oscillating laser-MIG hybrid welding of thick aluminum alloys was conducted by Yang *et al.* [29]. The results showed that the porosity decreased by utilizing oscillating laser, as oscillating laser promoted the bubbles in the molten pool to be captured by laser induced keyhole. Jiang *et al.* [6] pointed that the pore-free weld could be obtained with high laser oscillating frequency because the effect of columnar grains growth on bubble interception disappeared. Besides, Wang *et al.* [30] found that the laser beam oscillation could increase the keyhole opening size and enhance the keyhole stability. Additionally, the turbulence was formed to reduce the possibility of bubbles moving towards the solidification front due to the beam stirring effect. Zhang *et al.* [31] discussed the relationship of keyhole velocity and porosity. The authors pointed out that the porosity defect could be suppressed, since the keyhole movement was stabilized due to the oscillating laser. Besides, a cyclical drag force on the keyhole-melt interface was generated due to the oscillating laser [32].

Many issues, i.e., arc deflection and keyhole instability need to be solved for narrow-gap laser-MIG hybrid welding of thick aluminum alloys, which is more complicated than titanium alloys and steel. Oscillating laser is an effective way to improve the welding characteristics for aluminum alloys. However, the influence law and mechanism of oscillating laser on weld formation and porosity defect are complicated and remain unclear. In view of this, oscillating laser was employed in narrow-gap laser-MIG hybrid welding of aluminum alloys to optimize the weld formation and inhibit pore defect in the present study. The influences of laser power, welding speed, oscillating frequency and diameter on weld formation and porosity were investigated. The arc behaviors with different laser oscillating parameters were analyzed and the relationship of arc behavior and welding formation was clarified. The keyhole dynamic behavior was observed and investigated. The relationship of keyhole dynamic behavior, Rayleigh jet instability and pore defect was discussed.

2. Experiment procedure

2.1. Materials and equipment

5A06 aluminum alloys with the size of 100 mm × 80 mm × 16 mm and ER5356 with the diameter of 1.2 mm were employed as base metals and welding wire, respectively. The composition of the base metal and filler wire are shown in Table 1. A 10 kW fiber laser (TRUMPF LASER TruDisk 10002) with continuous laser output mode were employed. The laser wavelength is 1070 nm and the focal distance is 400 mm. The welding machine (Fronius TransPuls Synergic 4000) was used and pulse MIG welding mode was adopted. The welding current and voltage were automatically matched with the wire feeding speed. The high-speed video (Phantom V2512) with the frame rate of 20,000 frames/s was utilized in the experiment. The spectrometer was employed to monitor the plasma behavior and calculate the plasma temperature and density.

Table 1. Chemical compositions of 5A06 aluminum alloys and welding wire.

Materials	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Al
5A06	≤0.40	≤0.40	≤0.10	0.50 ~ 0.80	5.80 ~ 6.80	–	≤0.20	0.02 ~ 0.10	Bal.
ER5356	≤0.25	≤0.40	≤0.10	0.05 ~ 0.21	4.50 ~ 5.50	0.05 ~ 0.20	≤0.10	0.06 ~ 0.20	Bal.

2.2. Experiment procedure

The schematic diagram of welding process is shown in Fig. 1. The U-shaped narrow-gap groove with the angle of 5° was machined on the base metals. Before welding, the base metals were brushed and cleaned with alcohol to get rid of oxidation and contamination. Laser leading welding mode was adopted. The angles between laser beam, arc torch and workpiece are 80° and 55°, respectively. The defocus distance of laser beam is 0 mm. The distance between laser beam and welding wire filler is 3 mm. Welding parameters, including laser power, welding speed, wire feeding speed, laser oscillating frequency (R_f) and diameter (R_d) are listed in Table 2. High-purity argon was employed as the shielding gas at a flow rate of 30 L/min. During welding process, the workpiece was clamped on a platform. The platform was controlled by stepper control system. High-speed video was set to monitor the arc behaviors and keyhole dynamic behaviors. “Sandwich” structure was used to monitor the keyhole dynamic behaviors, as shown in Fig. 1(b). The setup schematic diagram of high-speed video and spectrometer are shown in Fig. 1(a) and (b).

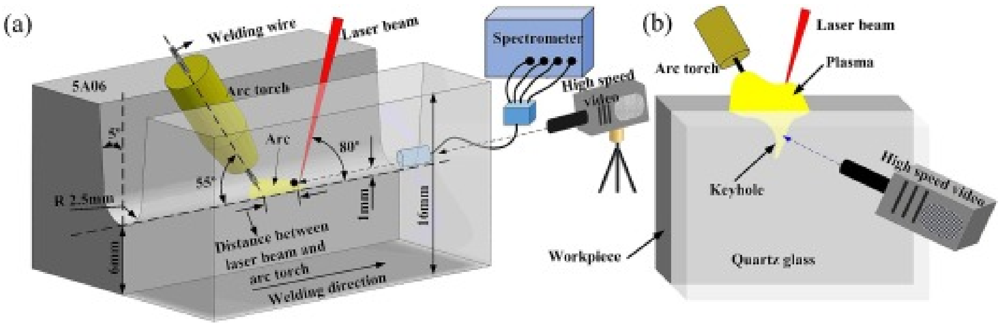

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Fig. 1. Schematic diagram of narrow-gap oscillating laser-MIG hybrid welding process.

Table 2. Oscillating laser-MIG hybrid welding parameters.

Numbers	Laser power /kW	Wire feeding speed /m/min	Welding speed /mm/s	Laser oscillating frequency /Hz	Laser oscillating diameter /mm
1	2	8	12	0	0
2	2.5	8	12	0	0
3	3	8	12	0	0
4	3.5	8	12	0	0
5	4	8	12	0	0
6	4.5	8	12	0	0
7	3.5	6	12	0	0

Numbers	Laser power /kW	Wire feeding speed /m/min	Welding speed /mm/s	Laser oscillating frequency /Hz	Laser oscillating diameter /mm
8	3.5	7	12	0	0
9	3.5	8	12	0	0
10	3.5	9	12	0	0
11	3.5	10	12	0	0
12	3.5	11	12	0	0
13	3.5	12	12	0	0
14	3.5	9	12	150	1
15	3.5	9	12	300	1
16	3.5	9	12	450	1
17	3.5	9	12	600	1
18	3.5	9	12	300	2
19	3.5	9	12	300	3
20	3.5	9	12	300	4

After welding, X-ray non-destructive testing equipment was used to detect the weld porosity. The metallographic specimens of the cross-section were cut, ground and polished, then corroded with Keller's reagent. The cross-section morphologies of welds were observed by metallographic microscope (ZEISS Axio Observer A1m). The welding formation parameters of sidewall penetration (H_P) were measured from the cross-section morphologies.

3. Results

3.1. Weld formation

Weld surface and cross-section morphologies of weld obtained by different laser powers are shown in Fig. 2(a)-(f). Metallic luster and fish-scale pattern were observed on all welds. Lack of fusion occurred in the weld obtained by lower heat input, as shown in Fig. 2(a). When the laser power varied from 2.5 kW to 4 kW, well-formed weld could be obtained, as shown in Fig. 2(b)-(e). When the laser power increased to 4.5 kW, the surface of the weld was unsmooth, as shown in Fig. 2(f). When the laser power exceeded 3.5 kW, the typical laser weld with large depth and sharp tip at the bottom zone which was defined as laser action zone could be clearly observed. Different penetration depths on both sidewalls of the groove were observed on all cross-section morphologies, as shown in Fig. 2(a)-(f). The variation of difference between both sidewall penetrations with the changing of laser power is exhibited in Fig. 2(g). The symmetry of the weld cross-section was directly reflected by the difference between both sidewall penetrations. The weld symmetry changed with various laser powers. When the laser power was 2 kW, 2.5 kW, 3.5 kW and 4 kW, the difference between both sidewall penetrations was small and the weld morphology at cross-section was symmetrical. When the laser power was 2 kW and 2.5 kW, lower heat input resulted in small sidewall penetrations which caused small difference between both sidewall penetrations.

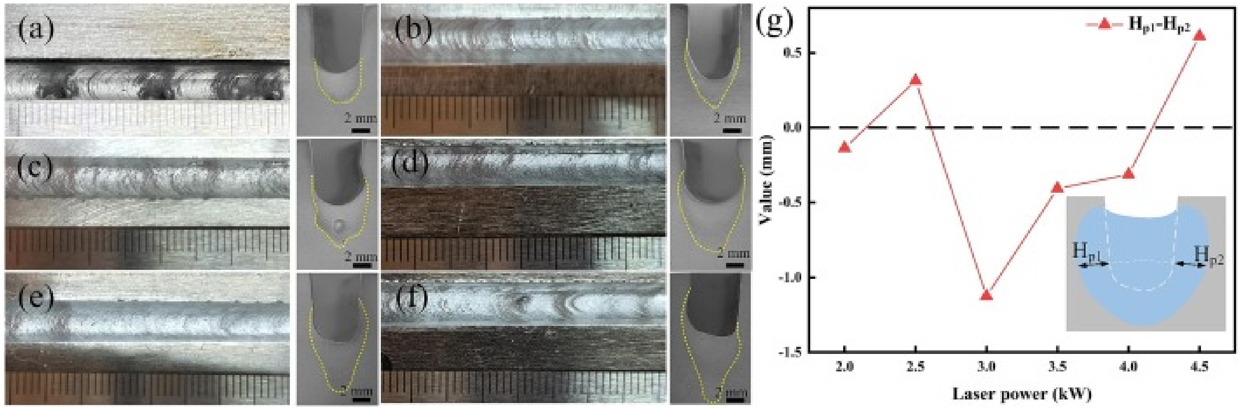

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Fig. 2. Weld surface and cross-section morphologies obtained by different laser powers (P) and corresponding difference between both sidewall penetrations: (a) P=2 kW, (b) P=2.5 kW, (c) P=3 kW, (d) P=3.5 kW, (e) P=4 kW, (f) P=4.5 kW, (g) difference between both sidewall penetrations.

Weld surface and cross-section morphologies obtained by different wire feeding speeds are exhibited in Fig. 3(a)-(g). When wire feeding speed varied from 6 m/min to 9 m/min, metallic luster and fish-scale pattern were observed on weld surfaces, as shown in Fig. 3(a)-(d). Laser action zones could be observed obviously on the cross-sections and the sidewall penetrations were small. Poor symmetry was observed on cross-sections of welds obtained by wire feeding speed of 6 m/min-9 m/min. The metallic luster was darked and fish-scale pattern disappeared on the weld surface obtained by larger wire feeding speeds of 10 m/min-12 m/min. Laser action zones could not be observed and arc action zones were more obvious on the weld cross-sections. The difference between both sidewall penetrations of weld obtained by different wire feeding speed is exhibited in Fig. 3(h). With the increase of wire feeding speed, the difference between both sidewall penetrations decreased. When the wire feeding speed was higher than 9 m/min, the difference between both sidewall penetrations was extremely small and the weld formation was symmetrical.

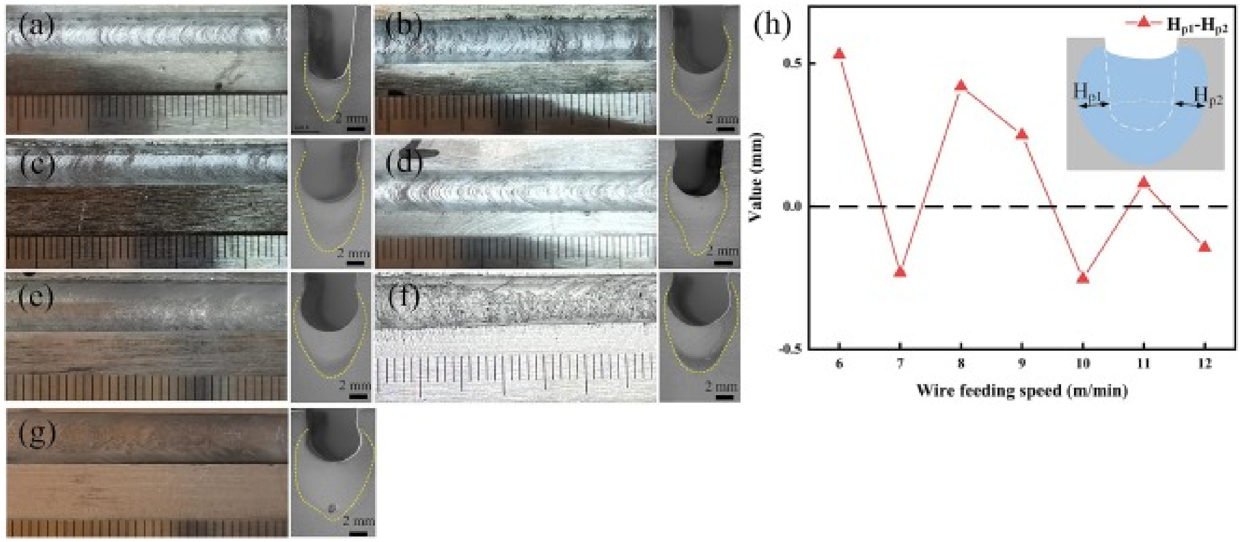

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Fig. 3. Weld Surface and cross-section morphologies obtained by different wire feeding speeds (V_f) and corresponding difference between both sidewall penetrations: (a) V_f=6 m/min, (b) V_f=7 m/min, (c) V_f=8 m/min, (d) V_f=9 m/min, (e) V_f=10 m/min, (f) V_f=11 m/min, (g) V_f=12 m/min, (h) difference between both sidewall penetrations.

Weld surface and cross-section morphologies obtained by different oscillating frequencies (R_f) are shown in Fig. 4(a)-(e). When oscillating laser-MIG hybrid welding was employed, metallic luster and fish-scale pattern were observed on all welds, as shown in Fig. 4(b)-(e). Oscillating laser was beneficial for the suppression of difference between both sidewall penetrations and little difference could be obtained while the laser oscillating frequency was 300 Hz and 450 Hz. The difference between both sidewall penetrations of oscillating laser-MIG hybrid welding is shown in Fig. 4(f). With the increase of oscillating frequency, the difference between both sidewall penetrations firstly increased and then decreased.

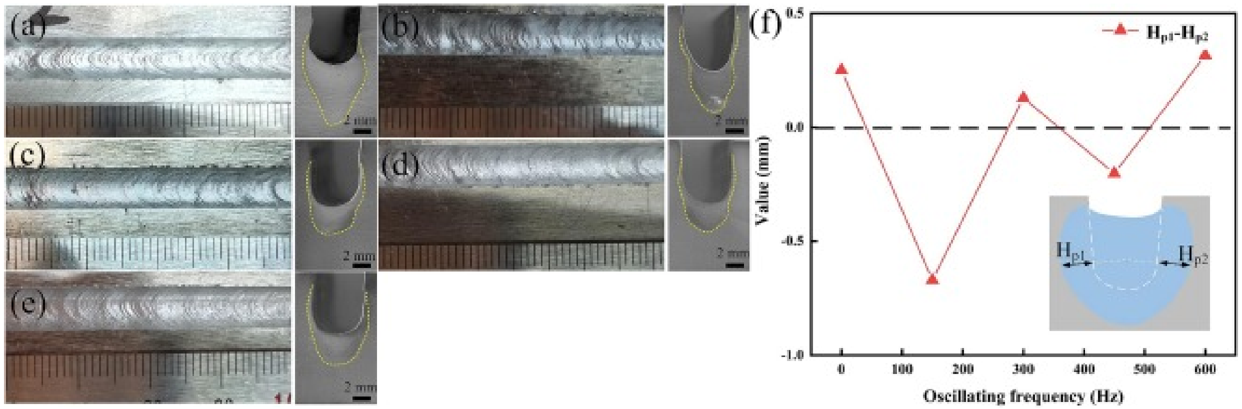

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Fig. 4. Weld surface and cross-section morphologies obtained by different laser oscillating frequencies: (a) R_f=0Hz, (b) R_f=150Hz, (c) R_f=300Hz, (d) R_f=450Hz, (e) R_f=600Hz, (f) corresponding difference between both sidewall penetrations.

Weld surface and cross-section morphologies obtained by different oscillating diameters (R_d) are shown in Fig. 5(a)-(e). When the oscillating diameters were higher than 2mm, the fish scale patterns on the weld surface were more uniform. The difference between both sidewall penetrations changed with oscillating diameter is shown in Fig. 5(f). With the increase of oscillating diameter, the difference between both sidewall penetrations firstly decreased and then gradually increased.

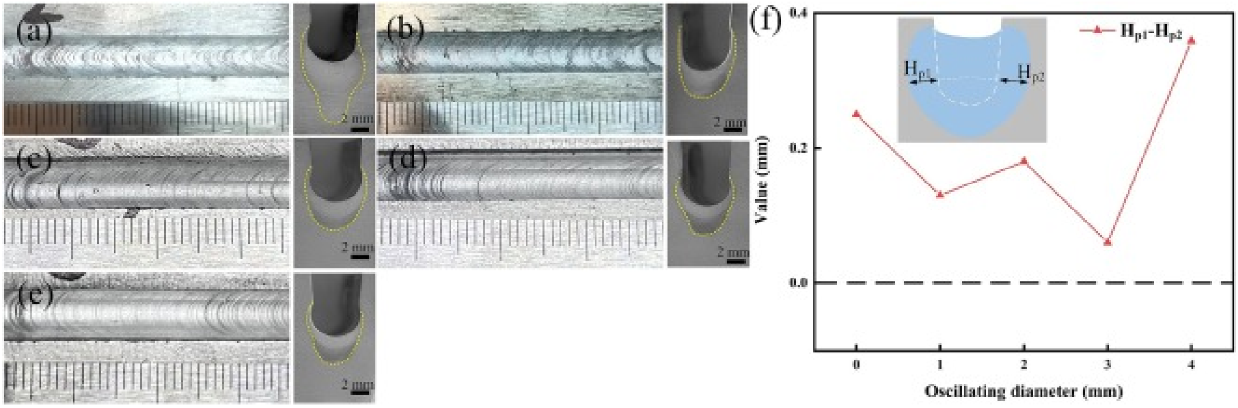

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Fig. 5. Weld surface and cross-section morphologies obtained by different laser oscillating diameters: (a) R_d=0mm, (b) R_d=1mm, (c) R_d=2mm, (d) R_d=3mm, (e) R_d=4mm, (f) corresponding difference between both sidewall penetrations.

3.2. Weld porosity

The influences of welding process parameters on weld porosities are shown in Fig. 6. The pores counted in the study are mainly process pores. The weld porosities obtained by different laser powers are shown in Fig. 6(a). With the increase of laser power, the porosity increased. The maximum porosity reached 3.5% in the weld obtained by maximum laser power of 4.5kW. When welded with high laser power, the keyhole depth increased and the keyhole stability decreased. During laser-MIG welding process, the keyhole closed periodically and the molten metal cooled quickly. The bubbles which formed at the bottom of the keyhole while keyhole collapsing, were captured by the solidification interface. The influence of wire feeding speed on weld porosity is shown in Fig. 6(b). When wire feeding speed increased, the porosity firstly increased and then decreased. The maximum porosity reached 4.1% in the weld obtained by wire feeding speed of 10m/min. While oscillating laser-MIG hybrid welding was adopted, the porosity decreased significantly, as shown in Fig. 6(c). With the increase of laser oscillating frequency, the weld porosity decreased. When laser oscillating frequency was higher than 300Hz, the porosities were maintained at an extremely low level and were all below 0.5%. Pore defects in welds obtained by different laser oscillating diameters and corresponding porosities are shown in Fig. 6(d). When laser oscillating frequency was 300Hz, the change of laser oscillating diameter had little effect on the weld porosity. The porosities of welds obtained by different laser oscillating diameters were all lower than 0.5%.

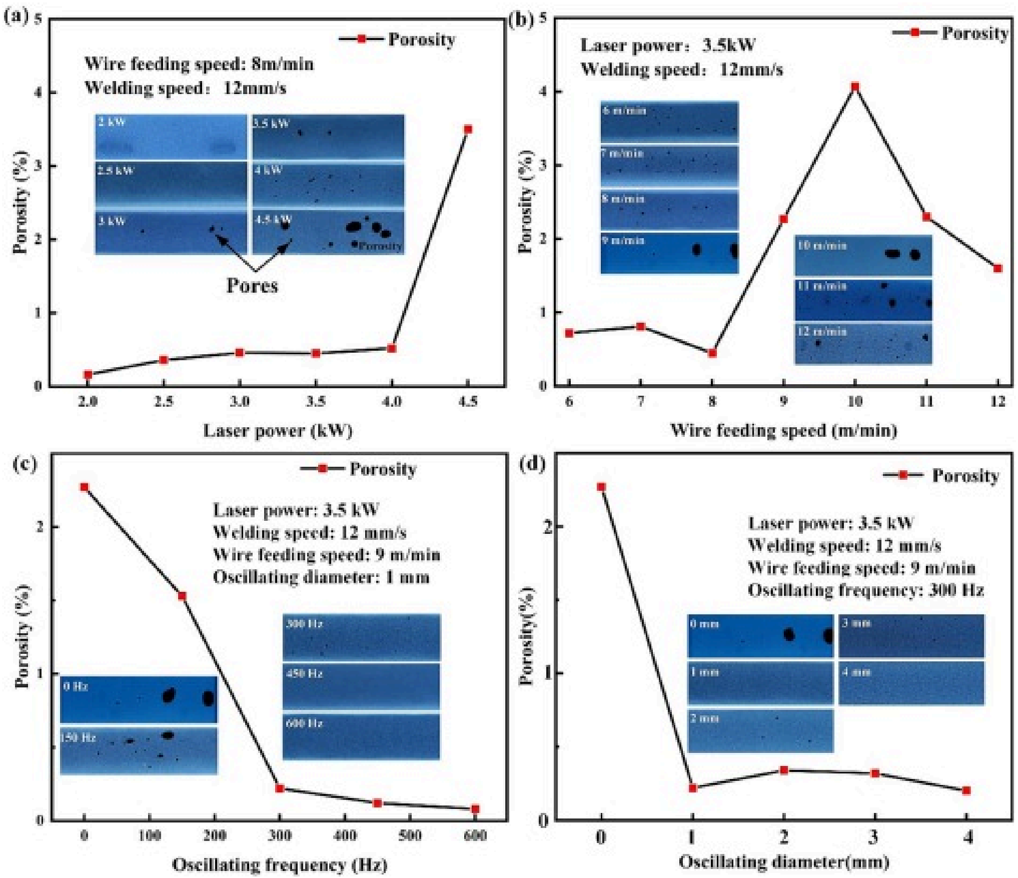

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Fig.6. Weld porosities varied with different welding parameters: (a) laser power, (b) wire feeding speed, (c) oscillating frequency, (d) oscillating diameter.

4. Analysis and discussion

4.1. Arc behavior with different welding parameters

The weld formation in the narrow-gap groove was influenced by arc behavior. The influence mechanism of arc behavior on the difference between both sidewall penetrations in narrow-gap groove is discussed and clarified. The arc behavior and deflection angle with different welding parameters are shown in Fig. 7. Six periods of arc were selected to characterize the arc behavior. The deflection angle was calculated by α_1-α_2, as shown in Fig. 7(e). The standard deviation (σ) of α was used to present arc fluctuation degree, which was calculated by Equation (1):

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(\alpha_i - \bar{\alpha})^2}{n}} \tag{1}$$

where α_i is arc deflection angle, $\bar{\alpha}$ is average value of arc deflection angle, n is sample quantity. Six periods of arc behavior obtained by low laser power of 2 kW and high laser power of 3.5 kW are shown in Fig. 7(a) and (b), respectively. At low laser power, the arc burnt unstably and was easily attracted by sidewall. The arc deflection angles and dispersion degree were large with low laser power, as shown in Fig. 7(a) and (e). The standard deviation of α was 23.7, as shown in Fig. 8. The arc deflection angles of high laser power decreased. The dispersion degree of arc deflection angle was smaller than that of low laser power, as shown in Fig. 7(e). The arc fluctuation degree of high laser power was about 37.6% of low laser power. At high laser power, the arc profiles were larger than that at low laser power. The heat input in the narrow-gap increased with the increase of laser power. More laser induced plasma was generated to maintain the burning of arc. A larger amount of metal vapor caused by higher laser power provided the conductive channel for stable burning of arc. The sidewall penetrations of welds obtained by low laser power were uneven, as shown in Fig. 2(a). During welding process at low laser power, the arc possessed a greater tendency to deflect towards one of the sidewalls and caused the difference between both sidewall penetrations. Higher laser power was conducive to stable burning of arc, resulting in a little difference between both sidewall penetrations, as shown in Fig. 2(d).

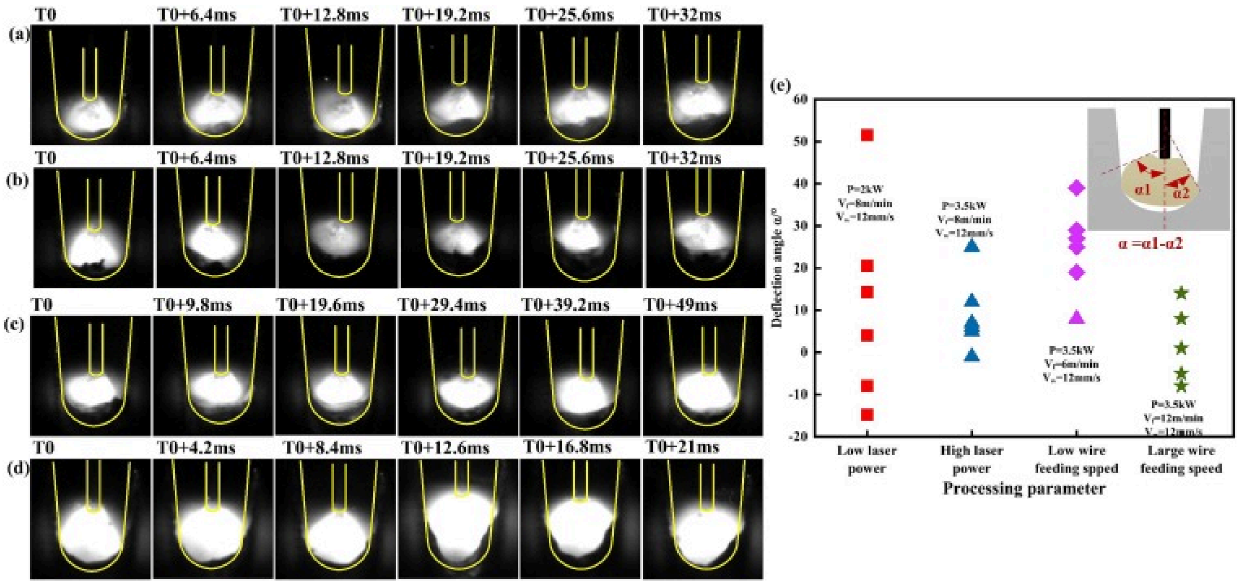

Fig. 7. Arc behavior and deflection angle of different welding parameters: (a) Low laser power (P=2kW, V_f=8m/min, V_w=12mm/s), (b) High laser power (P=3.5kW, V_f=8m/min, V_w=12mm/s), (c) low wire feeding speed (P=3.5kW, V_f=6m/min, V_w=12mm/s) (d) Large wire feeding speed (P=3.5kW, V_f=12m/min, V_w=12mm/s), (e) scatter diagram of arc deflection angles.

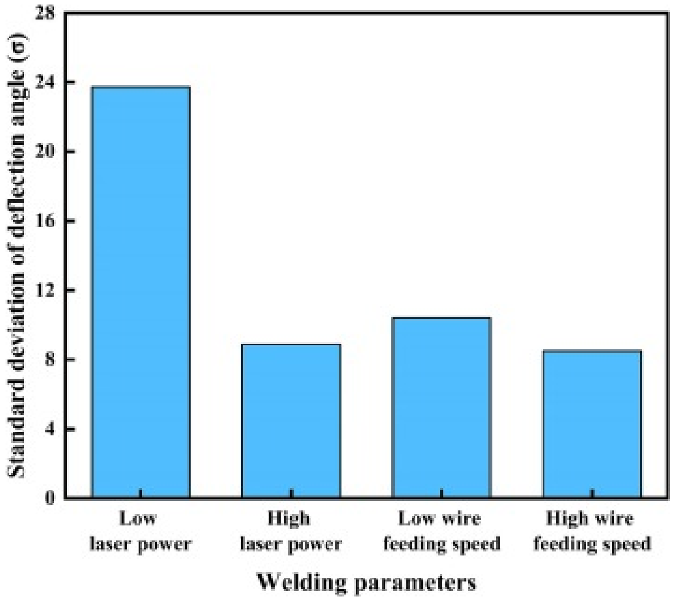

Fig. 8. The standard deviation of deflection angle obtained by different welding parameters.

Six periods of arc behavior obtained by low wire feeding speed and large wire feeding speed are shown in Fig. 7(c) and (d), respectively. The arc profiles of low wire feeding speed were small, as shown in Fig. 7(c). The burning arc was prone to being attracted by one of the sidewalls, as the current of low wire feeding speed was too small to maintain the stable burning of arc and resulted in the low stiffness of arc. The asymmetrical weld formation was obtained by low wire feed speed and the difference between both sidewall penetrations was large, as shown in Fig. 3(a) and (h). Compared with low wire feeding speed, the arc of large wire feeding speed burnt stably and the deflection angle of arc decreased significantly, as shown in Fig. 7(d)and (e). The arc fluctuation degree of large wire feeding speed was about 81.7% of that at low wire feeding speed, as shown in Fig. 8. During welding process of large wire feeding speed, arc was attracted by the electric channel and struck at the bottom of the groove rather than at the sidewalls. This illustrated that the difference between both sidewall penetrations obtained by large wire feeding speed was small, as shown in Fig. 3(g) and (h).

Arc behaviors obtained by different laser oscillating frequencies are displayed in Fig. 9(a)-(c). The arc behavior of laser-MIG hybrid welding with laser power of 3.5 kW, wire feeding speed of 9 m/min and welding speed of 12 mm/s without oscillating laser is shown in Fig. 9(a). The scatter diagram of arc deflection angles obtained with different laser oscillating frequencies is shown in Fig 0.9(d). The arc of oscillating laser-MIG hybrid welding burnt stably, as shown in Fig. 9(b) and (c). Compared with conventional laser-MIG hybrid welding, the deflection angle of arc obtained by laser oscillating frequency of 450 Hz decreased significantly and the fluctuation degree decreased by about 60.3%, as shown in Fig. 9(d) and Fig. 10. Little difference existed in the both sidewall penetrations of oscillating laser-MIG hybrid welding while laser oscillating frequency was 300 Hz and 450 Hz, since oscillating laser was conducive to stable burning of arc with optimal frequency, as displayed in Fig. 4. However, the fluctuation of arc deflection angles obtained by oscillating frequency of 600 Hz was more violent than that of 450 Hz, as shown in Fig. 9(d) and Fig. 10. Consequently, the difference between both sidewall penetrations obtained by oscillating frequency of 600 Hz was larger than that of 450 Hz.

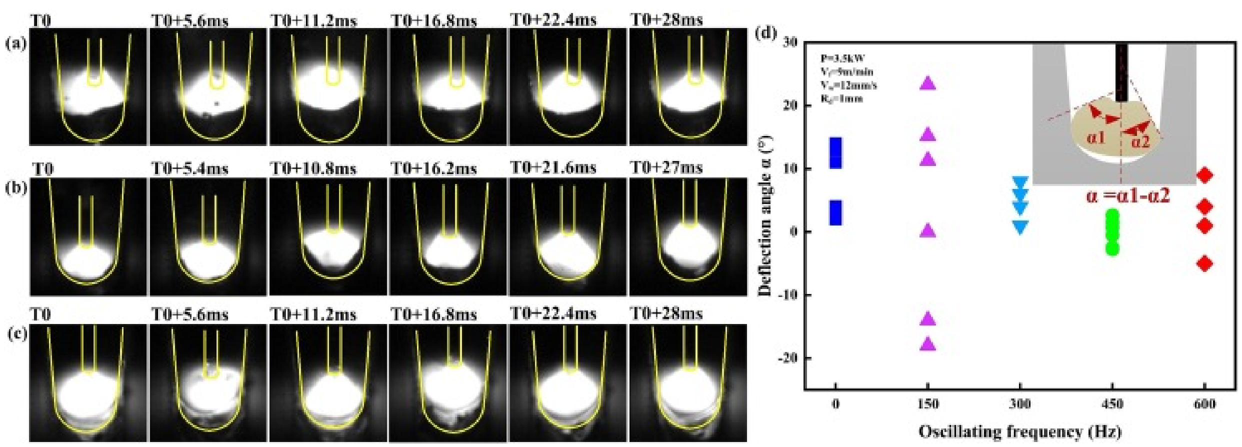

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Fig. 9. Arc behavior and deflection angle of different laser oscillating frequencies and corresponding deflection angle: (a) R_f=0Hz, (b) R_f=300Hz, (c) R_f=600Hz, (d) scatter diagram of arc deflection angles.

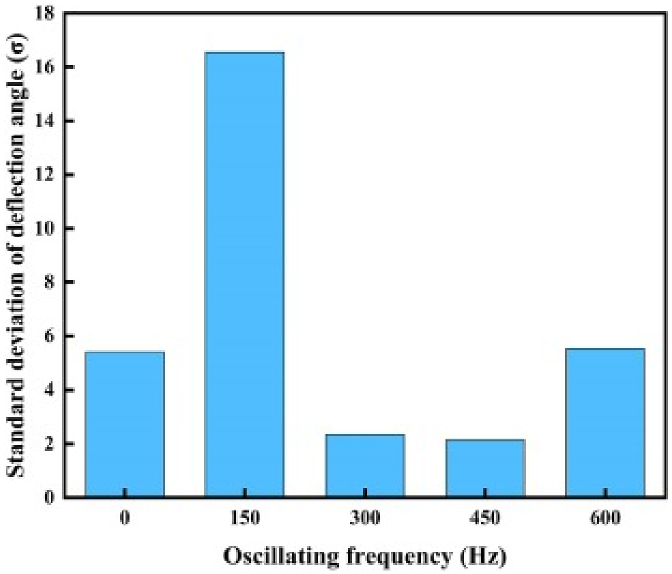

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Fig.10. The standard deviation of deflection angle obtained by different oscillating frequencies.

Arc behavior and corresponding arc deflection angles obtained by different laser oscillating diameters are shown in Fig. 11. With the increase of laser oscillating diameter, the arc deflection angle decreased firstly and then increased, as shown in Fig. 11(d). When laser oscillating diameter was 2mm, the arc was extremely stable and the arc profile was large, as shown in Fig. 11(b). Compared with conventional laser-MIG hybrid welding, fluctuation degree of arc decreased by about 58.9%, as shown in Fig. 12. When laser oscillating diameter increased to 4mm, the deflection of arc increased, as shown in Fig.

11(c). The arc deflection angles were dispersed and large, as the arc was attracted by laser beam. However, the welding filling heights of left and right sides obtained by oscillating diameter of 4mm were uniform, as shown in the section morphology of Fig. 5(e). The oscillating laser with large diameter facilitated the even spreading of molten metal in the groove. The increase of laser oscillating diameter caused the unstable burning of arc and the increase of difference between both sidewall penetrations, as shown in Fig. 5 (f). When laser oscillating diameter was 4mm, the laser beam would oscillate to the sidewall of the groove. Thus, the arc was attracted by laser beam and struck at the groove sidewall, which resulted in a large difference between both sidewall penetrations and the asymmetrical weld formation, as shown in Fig. 5(e).

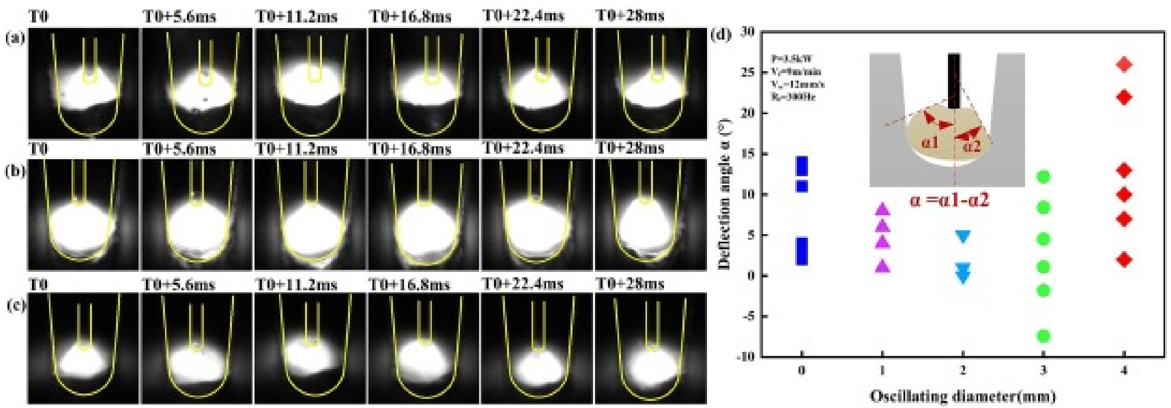

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Fig. 11. Arc behavior and deflection angle of different oscillating diameters and corresponding arc deflection angle: (a) R_d=0mm, (b) R_d=2mm, (c) R_d=4mm, (d) scatter diagram of arc deflection angles.

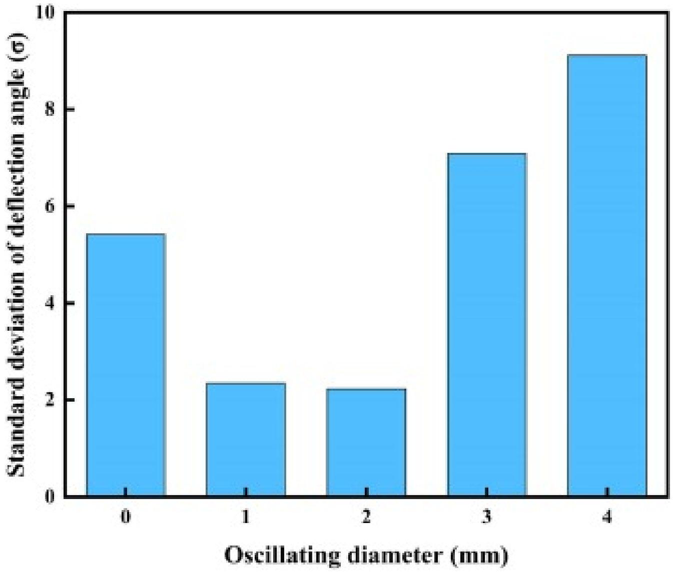

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Fig. 12. The standard deviation of deflection angle obtained by different laser oscillating diameters.

4.2. Arc combustion characteristic in narrow gap

In order to clarify the deflection behavior of arc, the spectral characteristics of plasma were analyzed. Typical plasma spectra of 5A06 aluminum alloys welded by different welding parameters are shown in Fig. 13. The spectrum lines including Mg I, Mg II and Ar II were concentrated among the range of 280nm-600nm.

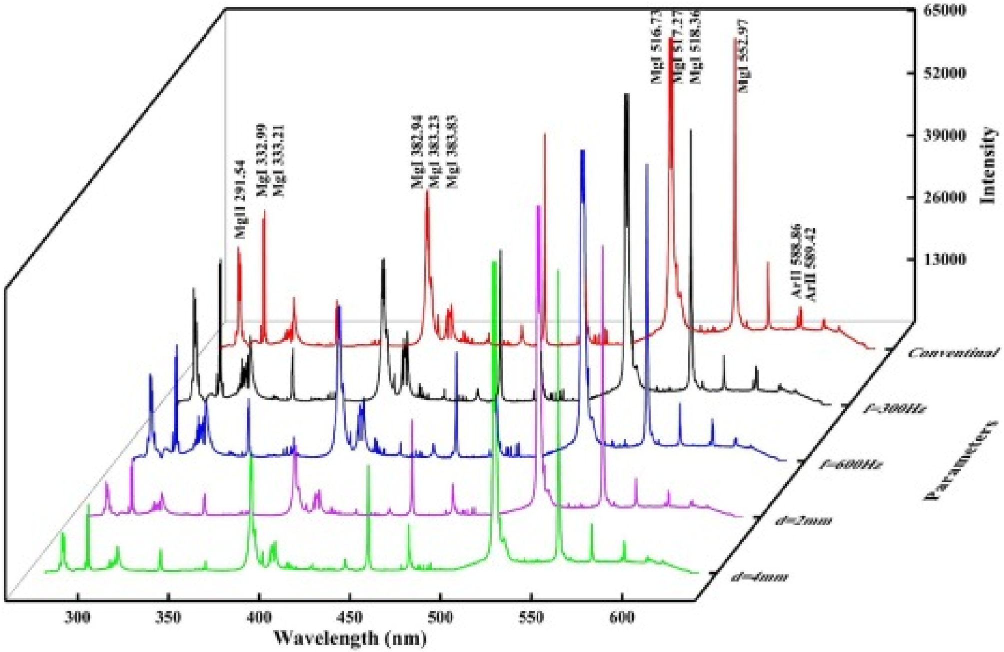

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Fig. 13. Typical plasma spectra with different welding parameters.

The Saha-Boltzmann plane method was utilized to calculate the plasma temperature. The selected characteristic spectral line parameters of Mg I are shown in Table 3. The plasma density was calculated using Lorentz fitting and Stark broadening.

Table 3. Spectrum line information selected by Boltzmann diagram.

Spectral line	λ(nm)	g	A(10^7/s)	E_i(ev)
Mg I	332.99	3	1.02	6.43
Mg I	333.21	3	1.7	6.43
Mg I	382.94	3	8.99	5.95

Spectral line	λ(nm)	g	A(10^7/s)	E_i(ev)
Mg I	383.23	3	6.74	5.95
Mg I	383.83	7	16.1	5.95
Mg I	516.73	3	1.13	5.12
Mg I	517.27	3	3.37	5.12
Mg I	518.36	3	5.61	5.12

Plasma temperature and density varied with laser oscillating frequency and diameter are shown in Fig. 14, respectively. When oscillating diameter was 1 mm, the plasma temperature was barely influenced by laser oscillating frequency, as shown in Fig. 14(a). The plasma density was significantly reduced when oscillating laser was employed. With the increase of oscillating frequency, the plasma density decreased. Compared with conventional laser-MIG hybrid welding, the plasma temperature and density decreased significantly when the oscillating diameter was higher than 2 mm, as shown in Fig. 14(b).

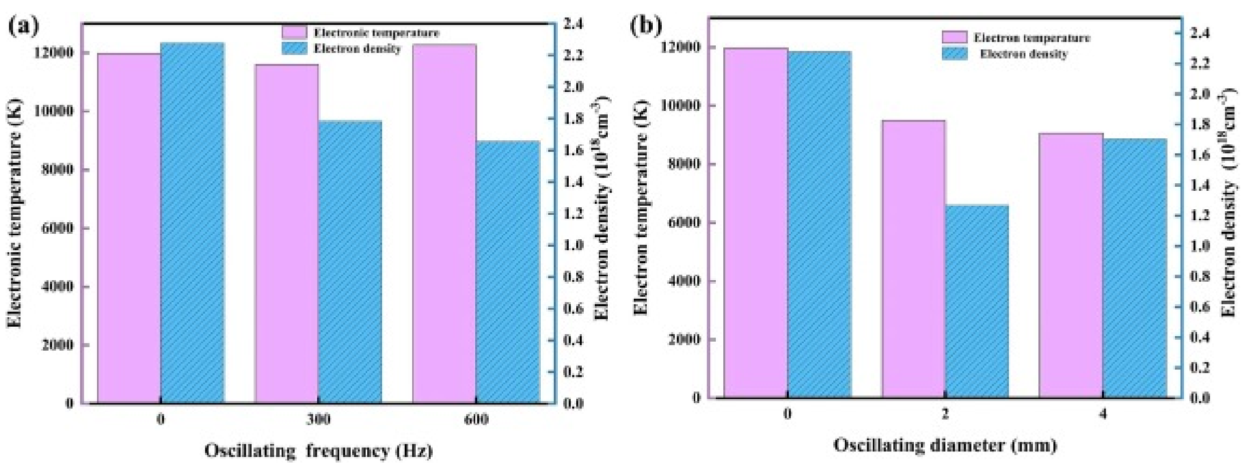

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Fig. 14. Plasma temperature and density obtained by different parameters: (a) laser oscillating frequency, (b) laser oscillating diameter.

The combustion characteristics of the arc are determined by the principle of minimum voltage, that is, by the volume and difficulty of the atmosphere to be ionized. The plasma resistivity is calculated by Equation (2):

$$\eta = \frac{\pi e^2 m_e^{1/2}}{(4\pi\varepsilon_0)^2 (k_B T_e)^{3/2}} \ln\Lambda \tag{2}$$

Where e is electron charge, m_e is electron mass, ε_0 is vacuum dielectric constant, k_B is Boltzmann constant, T_e is plasma temperature, $\ln\Lambda$ is Coulomb logarithm. In the study, the plasma temperature was about 12000K with different laser oscillating frequencies, as shown in Fig. 14a. The room temperature was approximately 300K. The resistivity ratio is shown in Equation (3):

$$\frac{\eta_{300K}}{\eta_{12000K}} = \left(\frac{12000}{300}\right)^{3/2} = 253 \tag{3}$$

The resistivity of room temperature gas is at least 253 times that of plasma produced by welding. High temperature plasma provided a conductive channel for arc striking. When arc struck at the cathode through the plasma conductive channel, the energy consumption for ionizing the surrounding atmosphere was the lowest. Schematic diagram of plasma distribution of different welding methods is shown in Fig. 15. During conventional laser-MIG hybrid welding process, metallic vapor erupted from the keyhole. Electrons produced by excitation were gathered at the keyhole opening, as shown in Fig. 15(a). Cathode emission electrons were gathered at the groove sidewalls, as shown in section A-A. The distance between filler wire tip and electrons produced by excitation was L_1, while that between filler wire tip and cathode emission electrons was L_2. As the electrons produced by excitation were gathered at the bottom of groove, $L_1 > L_2$. According to minimum voltage principle, the volume of ionized atmosphere required for arc striking between the filler wire tip and channel L_2 was smaller than that of L_1. Thus, the arc was attracted by the sidewall, as shown in Fig. 9(a).

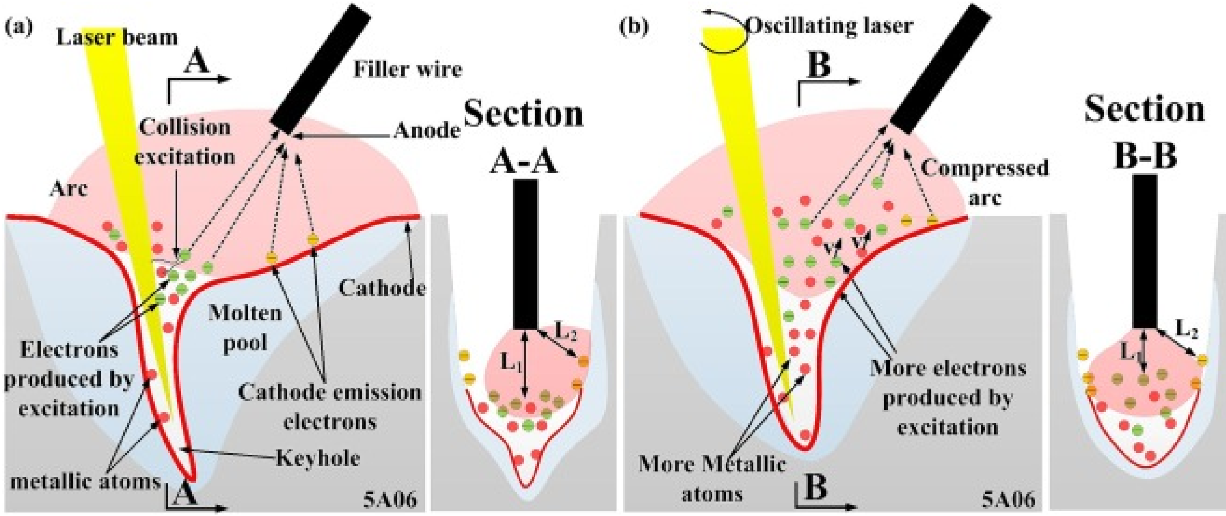

Fig. 15. Schematic diagram of the influence of plasma distribution on arc behavior: (a) conventional laser-MIG hybrid welding, (b) oscillating laser-MIG hybrid welding.

During oscillating laser-MIG hybrid welding process, keyhole opening size was larger and more metallic vapor erupted from the keyhole and collided with photons, as shown in Fig. 15(b). Compared with conventional laser-MIG hybrid welding, more electrons were produced by excitation. Under the action of the oscillating laser, electrons moved towards the upper part of the groove. The distance between filler wire tip and electrons produced by excitation was shorter, $L_1<L_2$, as shown in section B-B. According to minimum voltage principle, the arc struck at the center bottom of groove. The welding current and voltage fluctuation during conventional laser-MIG hybrid welding and oscillating laser-MIG hybrid welding process are shown in Fig. 16. The welding voltage of oscillating laser-MIG hybrid welding was generally lower than that of conventional laser-MIG hybrid welding, which supported the above analysis.

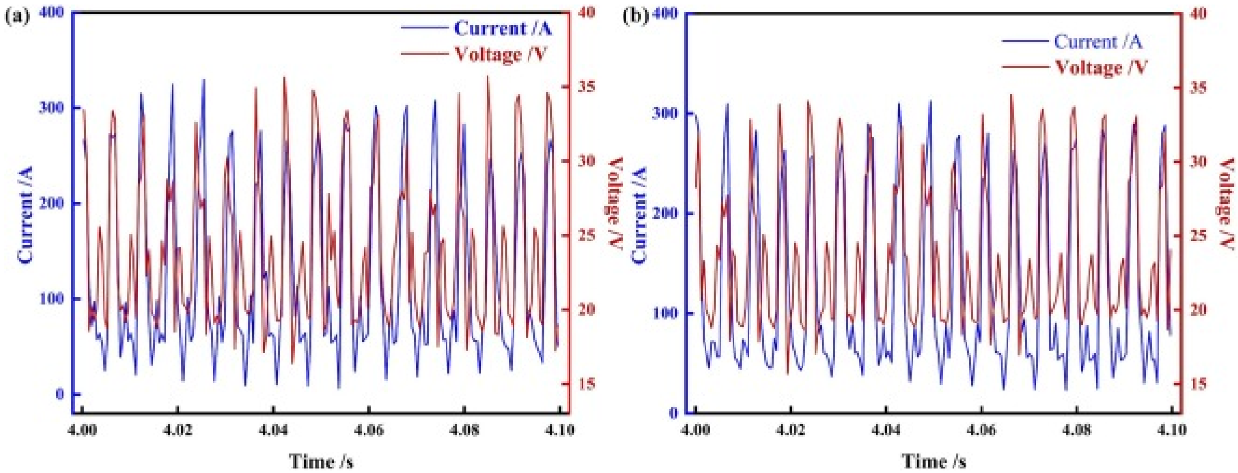

Fig. 16. The welding current and voltage fluctuation during welding process: (a) conventional laser-MIG hybrid welding, (b) oscillating laser-MIG hybrid welding (R_f=300 Hz, R_d=1 mm).

4.3. Keyhole dynamic behavior

Process pores in weld were closely related to the keyhole dynamic behavior. During deep penetration welding process, the collapse of keyhole wall resulted in the periodic closing of keyhole. The smaller the keyhole opening size and the deeper the depth, the harder for the formed bubbles to escape when the keyhole collapsed. During conventional welding process, deep and narrow keyhole resulted in higher porosity of 2.3%. During oscillating laser-MIG hybrid welding process, the keyhole was shallow and wide, which was conducive to the escaping of pores.

The keyhole behaviors of different welding parameters are shown in Fig. 17. The keyhole behaviors of conventional laser-MIG hybrid welding were unstable with laser power of 3.5 kW, wire feeding speed of 9 m/min and welding speed of 12 mm/s, as shown in Fig. 17(a). The keyhole depth fluctuated violently and varied from 4.3 mm to 11.8 mm. The keyhole opening size was small and varied from 0.8 mm to 1.4 mm. The keyhole behaviors of oscillating laser-MIG hybrid welding with oscillating frequency of 600 Hz and oscillating diameter of 1 mm were more stable than that of conventional welding, as shown in Fig. 17(b). The keyhole depth fluctuated slightly which varied from 6.8 mm to 10.1 mm. The keyhole opening size of oscillating laser-MIG hybrid welding was approximately twice the keyhole opening size of conventional welding. The keyhole opening sizes of oscillating laser-MIG hybrid welding were more stable and varied from 2.3 mm to 3.8 mm.

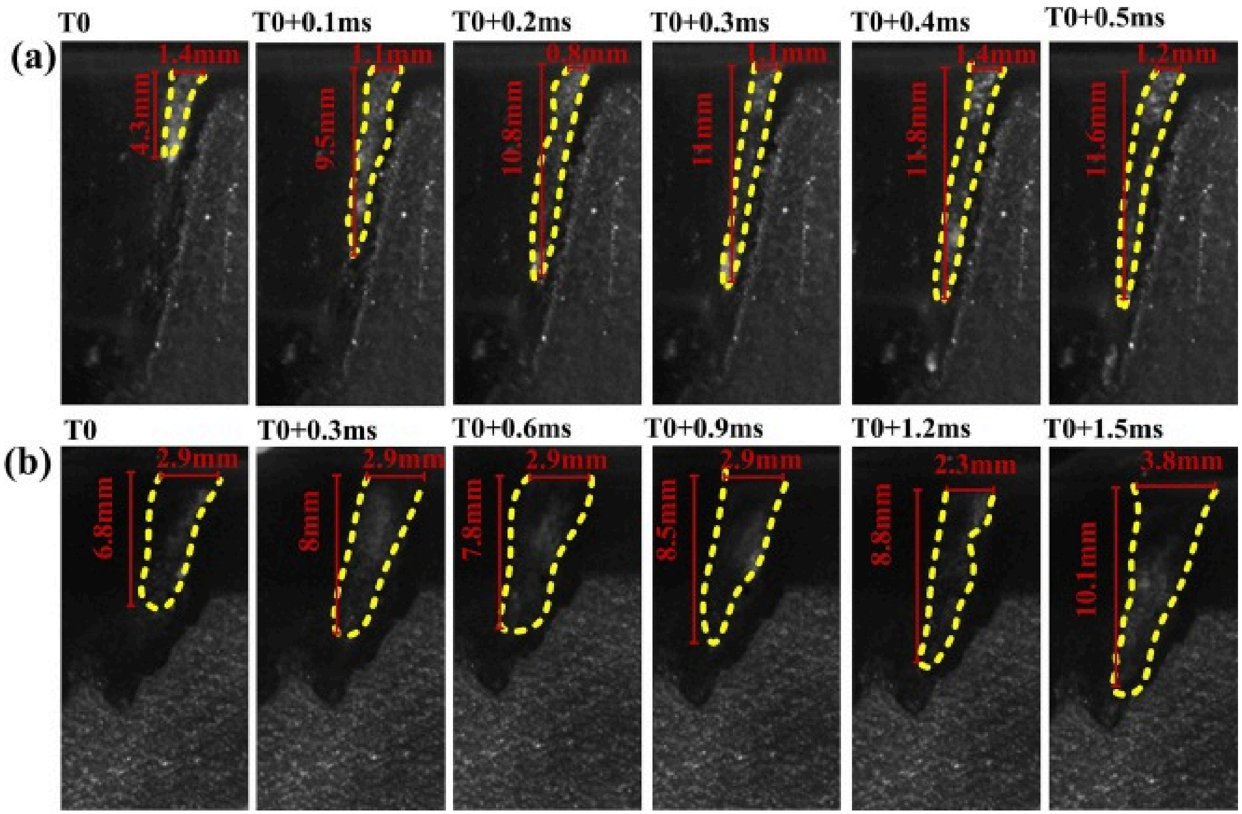

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Fig. 17. Keyhole dynamic behavior during welding process: (a) conventional laser-MIG hybrid welding, (b) oscillating laser-MIG hybrid welding.

The instability and collapse of keyhole were related to the bulge on keyhole wall, which was induced by Rayleigh jet instability [33]. The criteria for Rayleigh jet instability is expressed as equation (4) [34]:

$$\lambda > 2\pi R \tag{4}$$

In this case, the wavelength λ of the disturbance was identified by keyhole depth h, the cylinder radius R was identified by half of keyhole opening size w.

During laser-MIG hybrid welding process, in order to prevent the formation of bulge, the ratio of keyhole opening size to keyhole depth should meet the criteria:

$$\frac{w}{h} > \frac{1}{\pi} \tag{5}$$

During conventional laser-MIG hybrid welding process, the ratios of keyhole opening size to keyhole depth were varied from 1/13.5 to 1/8.4 and significantly lower than 1/π when t>T0 ms, as shown in Fig. 17(a). Rayleigh jet instability occurred on the keyhole wall of conventional laser-MIG hybrid welding. The schematic diagram of bulge formation process on keyhole wall induced by Rayleigh jet instability is shown in Fig. 18(a) and (b). During welding process, keyhole was formed by the recoil pressure of metal vapor plume, as shown in Fig. 18(a). the keyhole depth h was larger than the keyhole opening size w. Necking at the bottom of the keyhole was induced by Rayleigh jet instability, resulting in formation of bulge on the keyhole wall, as displayed in Fig. 18(b). The bulge was impacted by shearing stream induced by vapor plume. The keyhole was severely unstable and gas cavity was generated at the bottom of keyhole, as shown in Fig. 18(c). During oscillating laser-MIG hybrid welding process, the ratios of keyhole opening size to keyhole depth varied from 1/3.8 to 1/2.3. Except for T=T0+1.2 ms, the ratios were greater than 1/π. The probability of Rayleigh jet instability on the keyhole wall during oscillating laser-MIG hybrid welding process was suppressed.

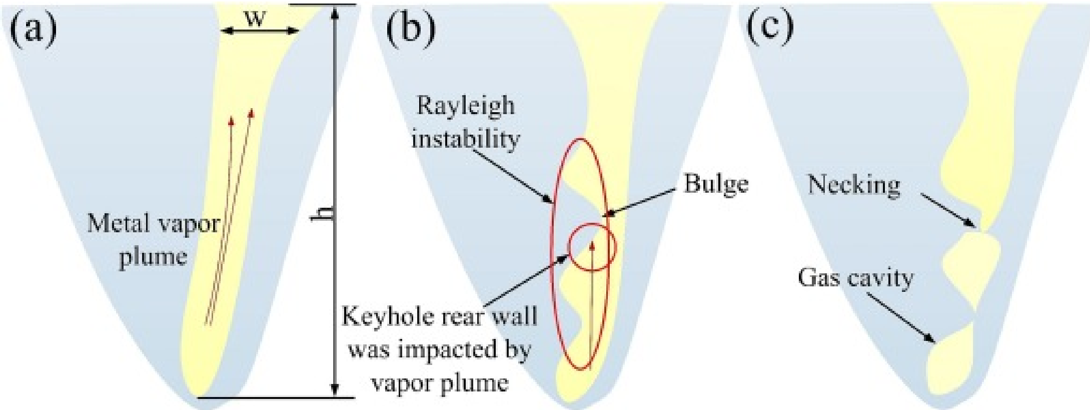

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Fig. 18. The schematic diagram of Rayleigh jet instability on keyhole wall: (a) keyhole was formed, (b) Bulge was induced by Rayleigh jet instability, (c) gas cavity was formed.

5. Conclusions

In the study, narrow-gap oscillating laser-MIG hybrid welding was employed for thick aluminum alloys. The relationship of weld formation and arc behavior was clarified. The influence of keyhole dynamic behavior on weld porosity was investigated. The conclusions were drawn as follows,

(1) The decrease of laser power, increase of wire feeding speed and applying of oscillating laser were conducive to the decrease of difference between both sidewall penetrations. The weld porosity was increased with the increase of laser power. When oscillating frequency was higher than 300Hz, porosity was decreased significantly.

(2) The both sidewall penetrations were uniform as the arc burnt stably and was struck at the center bottom of groove. Large wire feeding speed and oscillating laser facilitated stable burning of arc. Arc burnt stably and the

deflection angles were small while laser oscillating frequency was 300–450 Hz and diameter was 1–2 mm.

(3) With the action of oscillating laser, the excited electrons increased and diffused to the upper part of the groove. Arc was attracted by the excited electrons and struck at the center bottom of groove, which was conducive to inhibit the asymmetric weld formation.

(4) The weld porosity was related to the keyhole dynamic behavior. Compared with conventional laser-MIG hybrid welding, the probability of Rayleigh jet instability on the keyhole wall of oscillating laser-MIG hybrid welding was inhibited. The formation of bulge was suppressed and the stability of keyhole dynamic behavior was improved, which was conducive to the pore defects suppression.

CRediT authorship contribution statement

Chuang Cai: Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Jia Xie:** Writing – review & editing, Methodology, Investigation, Data curation. **Jie Yu:** Visualization, Software, Methodology. **Yonghong Liu:** Visualization, Methodology, Data curation. **Jiasen Huang:** Visualization, Data curation. **Hui Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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